PRINCETON UNIVERSITY

Princeton, NJ, 02/18/2026

Address: 35 Olden Street, 08540 Princeton, NJ
Email: cb7454@cs.princeton.edu
Re: ***Local Learning Rules for Out-of-Equilibrium Physical Generative Models***

Dear Dr. Curatolo,

We thank the Editorial Board and both Referees for their careful reading of our manuscript and for their constructive feedback. We have revised the manuscript substantially. The main changes (marked in red in the revised manuscript) are:

- Clarified the equilibrium-sampling (ES) benchmark and expanded the discussion of trapping/mixing behavior.
- Improved the presentation of Fig. 1 (including the colormap) and performed an additional hyperparameter search to improve sampling quality.
- Scaled up the MNIST demonstration to standard 28×28 images (without downsampling), updating Fig. 2 and the corresponding text accordingly.
- Improved notation and exposition, including clarifying the "hat" notation for energies/forces and simplifying the notation for the asymmetric coupling.
- Addressed Referee B's main concern regarding time-dependent parameters by clarifying that the learned parameter protocol is applied open-loop at inference time (no measurements or feedback), and by adding a brief discussion of feasible experimental platforms with time-dependent parameters.

We believe these changes have significantly improved both clarity and scope.

Sincerely,
Cyrill Bösch, Geoffrey Roeder, Marc Serra Garcia and Ryan Adams

Report of Referee A

The manuscript describes a novel way to train generative models in an out-of-equilibrium setting using local forces only. This is an interesting suggestion that might be relevant in the future for physical implementations of learning systems. The idea is interesting and timely, and I expect it will be of interest to the larger physics community.

We thank the Referee for the positive assessment.

In terms of presentation, the manuscript is overall clearly written though I find that some details are lacking, especially for readers who are not familiar with the inner workings of reverse SDEs (like myself).

I recommend publication in PRL with minor revisions, and would appreciate the authors taking these points in consideration:

1. The details of the "benchmark" process (SE) are not clear to me and there are no details in the SI. It is especially unclear why SE gets trapped in a single attractor.

We thank the reviewer for pointing out this omission. To benchmark the process, we simulate the Langevin dynamics of the system, recording the position every τ time-intervals. This has now been clarified in the manuscript.

The learned energy landscape of the two models (figs 1b and 1e) are identical, with a saddle at the origin, and since the initial distribution is a zero-centered Gaussian, I would assume that some of the initial points would fall into the the other basin…

We thank the Referee for pointing out this subtle point of confusion. For highly symmetric cases, it is indeed true that re-starting the Langevin dynamics multiple times at the origin would populate both wells. However, this simplifying assumption breaks even for small variations from this ideal landscape. For example, if there is an additional Gaussian peak/potential well at the origin, the system will not approximate the target distribution unless it overcomes the potential barriers enough times to equilibrate, which requires waiting multiple Kramers times. Since the goal of the article is to develop a method for general probability distributions, such simplifying assumptions cannot be made. We included a discussion of this matter in the manuscript.

… or at least that this could be fixed by changing the temperature. (I assume this question reflects a basic misunderstanding on my part, but I'd appreciate some clarification from the authors.)

Changing the temperature will not preserve the equilibrium probability (Boltzmann) distributions, causing the distribution to broaden/narrow, unless the parameters are also changed (in which case, the systems time to convergence will remain unchanged). Dynamically adjusting the temperature, for example, by progressively cooling the system, although intuitive, it is also known to not solve the exponential mixing time problem [1]. This is precisely the reason why diffusion models have been introduced.

[1] Robert, C. P., Casella, G., & Casella, G. (2004). *Monte Carlo statistical methods* (Vol. 2). New York: Springer.

2. In passing I'd remark that I find the choice of colors for the heatmaps in Fig 1 to be quite poor. It is very hard to discern the contours, both on screen and in a printed copy.

We thank the referee for this suggestion and have updated the color scheme of the heatmaps in Fig. 1 and the figure in the Supplemental Material to improve readability.

3. For the mixture-of-Gaussians experiment, I think the paper would benefit from a quantitative comparison of the sampled distributions. Looking at Fig 1a and 1d, I find two potentially interesting observations:

- While SE collapses to one mode, it seems that it better learns this mode's mean and standard deviation. In addition, it seems that SGM overestimates the mean of the positive modes and underestimates the mean of the negative one, as if there's a "force" pushing it away from the origin. It also seems (though it's hard to say by eye) that SGM consistently underestimates the standard deviation of the marginals.

- SGM generates a non negligible number of spurious samples in between the modes, where the actual density is very low. It would be interesting to quantify this and comment on how this depends on hyperparameters and training time (cf. Biroli et. al: https://arxiv.org/abs/2402.18491).

Are these observations correct? If so, do we understand them?

We thank the referee for these careful observations and the interesting reference.
We performed an additional hyperparameter search and found that operating at a slightly lower temperature significantly improves sample quality. At this temperature, the spurious samples between

the modes vanish. The means and variances of the modes are also captured more precisely. We have updated the figures accordingly with the new sampling results.

4. Some discussion is due about scalability, required compute, etc. Why is it necessary to downsample MNIST? Is an image of 28x28px a challenging task for this method? If so, are there any avenues towards making it work in non-toy scenarios?

We have updated the code to train and sample full-resolution 0-1-MNIST (28×28) and changed Fig. 2 accordingly. The earlier downsampling was used only for rapid prototyping and is not a fundamental limitation. Moving beyond MNIST is a natural direction but beyond the scope of this Letter, which is intended to provide the theoretical foundation.

5. Very minor: The authors say that they force the parameter gamma to be positive. I'm curious about whether this constraint is actually enforced in practice. That is: does the system tends to evolve towards negative gamma otherwise?

In our experiments the learned parameters typically satisfy $\gamma_n>0$ without any explicit constraint: the optimization consistently drives γ_n to positive values. Occasionally, when a particular γ_n is weakly identified (effectively fluctuating around zero due to noise/under-training), it can become slightly negative. Because a single negative γ_n renders the corresponding energy contribution non-confining and can destabilize the reverse-time integration, we enforce $\gamma_n >0$ in the implementation. We have clarified this point in the Supplemental Material.

Report of Referee B

The authors present a method for training physical systems to perform generative tasks by developing local learning rules for out-of-equilibrium diffusion-based models. They show that the driving protocol of a score-based generative model (SGM) can be learned directly from local force measurements, without requiring global optimization or equilibrium assumptions. Using a network of nonlinear, overdamped oscillators coupled to a thermal bath, they demonstrate sampling from a two-dimensional Gaussian mixture and, at larger scale, image generation on a 12 × 12 oscillator array trained on downsampled MNIST digits "0" and "1."

This is an interesting paper that bridges physical learning and generative modeling, and will be of interest to physicists working in these areas. However, I do not believe it meets the broad-applicability requirement for Physical Review Letters, because the proposed scheme produces time-dependent model parameters θ(t). For implementation in a real physical system, parameters must be fixed in time so that the system's natural dynamics alone perform the computation. The approach presented here would require an external controller continuously reprogramming the parameters of a device during operation, meaning that the computation is not carried out by the physical dynamics of the device. This restriction means that the scheme is not a viable way of training real physical devices to do computation.

We would like to respectfully disagree with the referee's assessment. Firstly, time-dependent parameters do not require any feedback. They have been realized in fully passive waveguide systems where time evolution is implemented with an additional space dimension [1,2]. Secondly, while, of course, eliminating time-dependence would simplify the realization, this is precisely the fundamental characteristic that allows diffusion models to outperform equilibrium sampling methods such as the well-studied Boltzmann machines (BMs). And, thirdly, systems with time-dependent parameters are a very active field of research, displaying rich phenomena ranging from time boundaries [3,4] to nonreciprocity [5-7], temporal pumping [8], Floquet bands [9] and physical learning beyond the quasistatic limit [10]. (Thus, the focus on time-dependent problems makes this research highly timely and of broad interest). The significance of our results lies in demonstrating that the principles

underpinning learning in physical systems [10-12] and the brain [13,14] can be generalized to such systems with time-dependent out-of-equilibrium dynamics and applied to generative models.

We clarify these points in the manuscript on page 2 and explicitly distinguish open-loop parameter protocols from feedback control, to avoid the impression that inference relies on an external controller performing state-dependent computation. We have also extended the introduction by clarifying the relation to BMs and added references to passive waveguides and systems with time-dependent parameters.

[1] Rechtsman, M. C., Zeuner, J. M., Tünnermann, A., Nolte, S., Segev, M., & Szameit, A. (2013). Strain-induced pseudomagnetic field and photonic Landau levels in dielectric structures. *Nature Photonics*, *7*(2), 153-158.

[2] Mukherjee, S., & Rechtsman, M. C. (2020). Observation of Floquet solitons in a topological bandgap. *Science*, *368*(6493), 856-859.

[3] Kim, B. L., Chong, C., & Daraio, C. (2024). Temporal refraction in an acoustic phononic lattice. *Physical Review Letters*, *133*(7), 077201.

[4] Wang, S., Shao, N., Chen, H., Chen, J., Qian, H., Wu, Q., ... & Huang, G. (2025). Experimental realization of temporal refraction and reflection in elastic beams. *Nature Communications*, *16*(1), 9520.

[5] Brandenbourger, M., Locsin, X., Lerner, E., & Coulais, C. (2019). Non-reciprocal robotic metamaterials. *Nature communications*, *10*(1), 4608.

[6] Ghatak, A., Brandenbourger, M., Van Wezel, J., & Coulais, C. (2020). Observation of non-Hermitian topology and its bulk–edge correspondence in an active mechanical metamaterial. *Proceedings of the National Academy of Sciences*, *117*(47), 29561-29568.

[7] Veenstra, J., Gamayun, O., Guo, X., Sarvi, A., Meinersen, C. V., & Coulais, C. (2024). Non-reciprocal topological solitons in active metamaterials. *Nature*, *627*(8004), 528-533.

[8] Xia, Y., Riva, E., Rosa, M. I., Cazzulani, G., Erturk, A., Braghin, F., & Ruzzene, M. (2021). Experimental observation of temporal pumping in electromechanical waveguides. *Physical Review Letters*, *126*(9), 095501.

[9] Park, J., Cho, H., Lee, S., Lee, K., Lee, K., Park, H. C., ... & Min, B. (2022). Revealing non-Hermitian band structure of photonic Floquet media. *Science advances*, *8*(40), eabo6220.

[10] Stern, M., Dillavou, S., Miskin, M. Z., Durian, D. J., & Liu, A. J. (2022). Physical learning beyond the quasistatic limit. *Physical Review Research*, *4*(2), L022037.

[11] Lopez-Pastor, V., & Marquardt, F. (2023). Self-learning machines based on Hamiltonian echo backpropagation. Physical Review X, 13(3), 031020.

[12] Stern, Menachem, et al. "Physical networks become what they learn." *Physical Review Letters* 134.14 (2025): 147402.

[13] Hopfield, John J. "Neural networks and physical systems with emergent collective computational abilities." *Proceedings of the national academy of sciences* 79.8 (1982): 2554-2558.

[14] Hebb, D. O. (2005). The organization of behavior: A neuropsychological theory. Psychology press.

I would also strongly recommend that the authors either replace Fig. 2 with one generated using standard 28 × 28 MNIST digits, or clearly qualify in the abstract and main text that their model reproduces a downsampled version of MNIST, rather than the standard dataset.

We thank the Referee for this suggestion. We have now scaled up the implementation to standard 28x28 0-1-MNIST (without downsampling) and have updated Fig. 2 and the corresponding text accordingly.

Smaller points: it would be helpful to more clearly state the purpose of the hat (caret) as it appears over the energy symbol. Similarly for \chi, the asymmetric couplings — it's not immediately clear why one has a hat and one doesn't.

We thank the referee for pointing this out. We have rewritten the paragraph below Eq. (7) to clarify the meaning of the hats used for the energies and forces. To avoid confusion, we have removed the hat from the asymmetric coupling and instead introduced a distinct variable, ω. These changes improve notational clarity and consistency throughout the manuscript.